

LESSON 107 (1.0 GROUND and 1.0 AATD)

Lesson Objective

During this lesson, the applicant will learn how to taxi and practice straight and level flight techniques in the training device. The applicant will also review the effects of controls. The applicant will also review previously learned maneuvers for proficiency.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on the basic elements of principles and aerodynamics of flight.
 - Quiz the student on the readings as needed.
 - Review chapter 03 of the Airplane Flying Handbook - Basic Flight Maneuvers:
 - The four fundamentals.
 - Effects and use of flight controls.
 - Attitude flying.
 - Straight and level flight.
 - Review chapter 7 of the Airplane Flying Handbook - Ground Reference Maneuvers:
 - Drift and Ground Track Control.
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 03 - Basic flight maneuvers.
 - Chapter 07 – Ground Reference Maneuvers
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 06 - Flight Controls.
- Warrior III Pilot's Information Manual.
 - Section 07 - Description and operation of the airplane and its systems.

Instructor Guidance

- This is the student's second simulator lesson and their first time practicing taxiing. Keep taxi expectations realistic — most students struggle with rudder pedal sensitivity and centerline tracking initially. Patience here builds good habits early.
- Always brief every taxi with an airport diagram. Make this a non-negotiable habit from the very first taxi, even in the simulator. Students who skip this step in training will skip it in real operations.

- Reinforce the hand-on-throttle habit during all ground operations. This is easy to establish now and critical to reinforce before the student gets to the aircraft.
- When introducing straight and level flight, resist the urge to demonstrate too much. Give the student the "Pitch + Power = Performance" framework, then let them feel the aircraft respond. Correct deviations through coaching rather than taking the controls.
- The "set pitch, then power, then trim" sequence should be stated and repeated every time a power or configuration change is made during this lesson. Repetition here prevents bad habits from forming later.
- When teaching straight and level at selected airspeeds, walk the student through the 100 RPM = 5-knot rule before demonstrating it, then have them predict what will happen before each power change. This builds aeronautical reasoning rather than passive observation.

Lesson Content (Ground)

Taxiing

- Introduce the before taxi check.
- Hand stays on the throttle.
- Heels on the floor (move feet up on the pedals when braking).
- Assist with wind corrections.
- Demonstrate with instrument checks.
- Assist with centerline and speed.
- Assist with runup.
- Ensure that every taxi is briefed with an airport diagram (except ramp).

Straight and Level Flight Techniques

- Pitch + Power = Performance.
- Set pitch THEN power THEN trim.
- Straight and level sight picture + 2300 RPM = 90KTS straight and level.
- Utilize the HSI and inclinometer to maintain heading.
- Utilize the Altimeter and VSI to maintain altitude.

Tracking a Ground Based Reference Line

- Go over Chapter 7 (Ground Reference Maneuvers - Drift and Ground Track Control) of the Airplane Flying Handbook.
- Introduce tracking a straight-line ground reference.
- Discuss the effects of wind.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic when weather permits.
- FAR regulation 91.111 - operating near other aircraft.
 - Cannot operate near another aircraft to create a collision hazard.
 - Cannot fly in formation unless there was a prior agreement between the PIC's of participating aircraft.
 - No passengers for hire in formation flight.
- Explain how to scan for traffic in daytime conditions.
 - 10 degrees per second, looking both above and below aircraft level.
- Clearing procedures:
 - During taxi.
 - Before climbs.
 - During descents.
 - Clearing turns.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Lesson Content (AATD)

Checklist Procedures

- Re-emphasize the importance of using checklists.
- Have the student run through all of the checklists.
- Assist if needed.
- Have the student go through the "passenger briefing" and the "engine fire during start brief".

Taxiing Procedures

- Introduce the before taxi check.
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- Assist with wind corrections.
- Demonstrate with instrument checks.
- Assist with centerline and speed.
- Assist with runup.
- Ensure that every taxi is briefed with an airport diagram (except ramp).

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- Demonstrate how to scan for traffic in daytime conditions.

- 10 degrees per second, looking both above and below aircraft level.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Review Effects of Controls as Appropriate

- If needed, refer to the lesson plan for 104.

Tracking a Ground Based Reference Line

- Use a straight-line ground reference and have the student track it.
- Ensure the student is correcting for wind drift.

Straight and Level Flight at Cruise Speed

- Demonstrate entering and maintaining straight and level flight.
- Pitch + Power = Performance.
- Set pitch THEN power THEN trim.
- Straight and level sight picture + 2300 RPM = 90KTS straight and level.
- Utilize the HSI and inclinometer to maintain pitch.
- Utilize the Altimeter and VSI to maintain altitude.

Straight and Level Flight at Selected Airspeeds

- For changes greater than 20 knots:
 - Full power or idle power.
 - Anticipate by 5 knots.
- For changes less than 20 knots:
 - 100 RPM change = 5 knots change.

Straight and Level Flight with Flaps

- Demonstrate the effects of flaps 10, 25 and 40 degrees.

Adding Flaps:

- Lift → Increases.
- Nose → Goes up.
- Elevator (initially) → Forward.
- Drag → Increases.
- Airspeed → Decreases.
- Elevator (due to drag) → Backward.

Removing Flaps:

- Lift → Decreases.
- Nose → Goes down.
- Elevator (initially) → Backward.
- Drag → Decreases.
- Airspeed → Increases.
- Elevator (due to drag) → Forward.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 108.
- Assign homework:
 - Review chapter 7 of the Airplane Flying Handbook - Ground Reference Maneuvers:
 - Drift and Ground Track Control.
 - Review checklists and briefings.

Completion Standards

This lesson is complete when the student can:

1. Have a weather briefing and a completed TOLD card prepared.
2. Demonstrate proper sequential checklist procedures for engine start, before taxi, ground check, before takeoff, and runway entry.
3. Correctly perform previously learned maneuvers with little instructor guidance.
4. Demonstrate an understanding of straight and level flight techniques in the training device.
5. Explain the relationship between power, airspeed, and angle of attack.